

Custom Build Brompton — Planning Document

Prepared for Asri · August 2026 · All prices in SGD equivalent, gathered from live listings/forums as of Aug 2026 — treat as indicative, confirm before ordering.

1. Requirements Recap

- Front and rear child seat capability required.
- Weight saving is a priority.
- Cost effectiveness is a priority.
- You travel for work — can buy overseas (incl. China) and collect/ship to a hotel.
- All costs quoted in SGD equivalent; tourist GST/tax rebates and Singapore import GST NOT factored in.

2. Executive Summary & Recommendation

Three viable paths were evaluated: (A) buy a used genuine Brompton (16" wheel) and modify it with titanium parts for weight savings; (B) self-build a 20" titanium Brompton-clone from a Chinese frame kit; or (C) buy a used genuine Brompton G Line (20" gravel-oriented wheel, launched 2024) and modify it with titanium + carbon parts. The table below summarises the trade-off.

Criteria	Option A: Used 16" Brompton + Ti mod	Option B: 20" Ti clone self-build	Option C: Used G Line (20") + Ti/carbon mod
Total cost (parts, excl. tools)	~S\$2,650 – 2,900	~S\$2,750 – 3,300	~S\$4,100 – 4,700
Bare-bike weight (both seats removed)	~10.3 – 10.6 kg	~8.5 – 9.5 kg	~11.5 – 12.5 kg
Child-seat compatibility	High — genuine Brompton geometry, all Brompton-fit seats/adapters work directly	Medium/Low — clone frame tube diameters & geometry vary by factory batch; brackets often need shimmming/custom fit	High — same head-tube/seat-tube standard as classic Brompton; Bromptonic/Milian/F4 seats explicitly list G Line compatibility
Build difficulty / tools	Moderate — swapping fork, rear triangle, seatpost on a known platform	High — full build from bare frame: wheelbuild, groupset install, brake/gear setup, alignment	Moderate-High — same swap logic as Option A, but G Line's disc brakes/Alfine hub and internal cable routing add a step
Risk	Low — well-trodden mod path, resale value retained (genuine Brompton)	Higher — frame quality varies by factory batch, no warranty, IP/trademark grey area, harder resale	Low-Moderate — genuine frame retains resale value, but G Line-specific Ti/carbon parts are a newer, thinner secondary market than classic Brompton parts
Fold compatibility (train/plane/hotel storage)	Identical to stock Brompton fold	Usually mimics Brompton fold but tolerances can be looser — verify before buying	Identical to stock G Line fold — bulkier folded package than classic Brompton due to 20" wheels + gravel tyres
Ride quality / use case	City/commuter, smooth-road only	Depends entirely on build spec	Best all-rounder — bigger wheels & tyres handle rougher roads/park connectors/light

Criteria	Option A: Used 16" Brompton + Ti mod	Option B: 20" Ti clone self-build	Option C: Used G Line (20") + Ti/carbon mod
			gravel, most reassuring platform for carrying kids

Recommendation: Option A (used 16" Brompton + titanium fork/rear-triangle/seatpost mod) remains the best cost-effective pick, but Option C (used G Line + Ti/carbon mod) is worth the premium if ride comfort and stability while carrying two children matters more to you than minimising cost.

- Option A wins on pure cost-effectiveness and weight-per-dollar, and has the deepest, cheapest secondary market for both the base bike and titanium parts.
- Option C costs roughly S\$1,400-1,800 more than Option A (mainly because the G Line itself is a newer, pricier used bike, and its disc-brake/wider-hub titanium parts are a smaller, costlier niche than classic Brompton Ti parts) — but it's still genuine Brompton, still keeps full child-seat compatibility, and its bigger 20" wheels/tyres and disc brakes make it noticeably more confidence-inspiring for carrying kids over uneven pavement, curbs, and park paths versus the classic 16" wheel.
- Treat Option B (20" titanium clone) as a secondary/future project once you've tested a mod path — build it as a second, ultralight bike for solo commuting once the kids have outgrown seats, not as the child-seat bike.

3. Option A — 16" Used Brompton + Titanium Modification

3.1 Bill of materials & cost (SGD equivalent)

Item	Est. cost	Source / notes
Used Brompton, steel M/A/C-type, 3-speed, good condition	S\$650 – 950	Carousell (SG, no shipping) is cheapest net-of-effort; Gumtree/Preloved UK ~£300-420 (~S\$520-730) is cheaper on paper but shipping a full bike overseas is costly/impractical — buy locally or during a UK trip and check it in as luggage
Titanium front fork (EasyTi or equivalent)	S\$260 – 380	Taobao direct via forwarding agent cheapest; eBay/EasyTi-brand resellers ~S\$460-520 (2-3x Taobao)
Titanium rear triangle	S\$450 – 650	Taobao direct cheapest; branded eBay listings ~S\$750-800
Titanium seatpost	S\$110 – 160	Taobao/AliExpress
Titanium bolt kit (full set, incl. hinge clamp bolts)	S\$90 – 130	Taobao/AliExpress
Lightweight tyres (e.g. Marathon Racer / Kojak, pair)	S\$80 – 100	Local bike shop or Taobao
Front child seat (mounted to head-tube block, e.g. Bobike Mini/Bromptonic-style)	S\$180 – 260	Taobao direct (S\$180-220) vs branded EU resellers (Bromptonic ~S\$260, Bobike+adapter ~S\$330)
Rear child seat + mount (e.g. Milian Pere-style, clamps to seat tube)	S\$230 – 330	Taobao direct (S\$230-280) vs Milian/SJS Cycles UK (~S\$330)
Pentaclip / seat-clamp compatibility hardware (if needed for rear seat)	S\$35 – 50	Brompton spares
Consumables (grease, cable, housing,	S\$40 – 60	Local

Item	Est. cost	Source / notes
misc.)		

Subtotal (parts): roughly S\$2,125 – 3,570, midpoint $\approx$ S\$2,650 – 2,900 once you settle on Taobao-sourced titanium and one seat brand at each end.

3.2 Weight plan

Configuration	Est. weight
Stock steel Brompton (3-speed, standard steel fork/rear triangle)	~11.2 – 11.5 kg
After: Ti fork (-0.35kg), Ti rear triangle (-0.45kg), Ti seatpost (-0.15kg), lighter tyres (-0.1kg)	~10.2 – 10.5 kg (bare bike)
+ Front child seat installed	+1.0 – 1.3 kg
+ Rear child seat + mount installed	+1.4 – 1.8 kg
Both seats fitted, ready to ride	~12.6 – 13.6 kg total

Note: child seats are removable/quick-detach on both ends, so day-to-day solo folding weight stays at the ~10.3kg bare-bike figure — most of the weight-saving budget is best spent on the fork and rear triangle since those give the biggest kg-per-dollar return.

3.3 Tools required

- Brompton-specific 2/2.5/3/4/5/6mm hex (Allen) key set + Torx T25/T30 (for hinge clamps and some Ti bolt kits)
- Torque wrench (2-20 Nm range) — critical for titanium bolts and hinge clamps, which are easy to over- or under-torque
- Pedal spanner / 15mm cone wrench
- Chain whip + cassette/freewheel removal tool (if swapping drivetrain parts)
- Bottom bracket tool (square-taper or splined, depending on BB type)
- Tyre levers + track pump
- Cable cutters, housing cutter
- Grease (standard + anti-seize/Ti-prep for titanium bolt threads — important to prevent galling)
- Bike repair stand (optional but strongly recommended for fork/rear-triangle swap)
- Estimated tool cost if starting from scratch: S\$150 – 260 (many items are one-off/reusable for future bikes)

4. Option B — 20" Titanium Brompton-Clone Self-Build

4.1 Bill of materials & cost (SGD equivalent)

Item	Est. cost	Source / notes
Titanium frame + fork + rear-triangle kit (20" trifold, Brompton-clone geometry)	S\$550 – 950	Taobao direct via agent is cheapest (factories: Hexati/EasyTi/"Comeplay" and similar OEMs); pre-assembled complete bikes from tibicycles.com (Singapore-based, TaiSeng) run S\$1,900 – 5,900 all-in, no self-build or shipping needed but 2-4x the DIY parts cost
Drivetrain: derailleur, shifter, crankset,	S\$300 – 450	Taobao/AliExpress or reuse a donor

Item	Est. cost	Source / notes
BB, chain, cassette/freewheel		groupset
Wheelset (20" alloy, hand-built or pre-built)	S\$180 – 400	Alloy cheapest; carbon wheelset adds S\$300-600+
Brakes (rim or disc, incl. levers)	S\$60 – 150	Disc adds cost & weight vs. rim
Handlebar (M/H/P/S-type equivalent), stem, grips	S\$60 – 120	
Saddle + Ti or alloy seatpost	S\$70 – 150	
Folding pedals	S\$40 – 90	
Front child seat + mount (expect custom shimming for clone tube diameter)	S\$180 – 260	+ S\$50-100 contingency for bracket fitting/fabrication
Rear child seat + mount (expect custom shimming)	S\$230 – 330	+ S\$50-100 contingency
Cables, housing, bolts, consumables	S\$80 – 120	

Subtotal (parts): roughly S\$1,850 – 3,470, realistic midpoint $\approx$ S\$2,750 – 3,300 including fitting contingency for the child-seat brackets.

4.2 Weight plan

Configuration	Est. weight
Ti frame + fork + rear triangle only	~1.6 – 2.0 kg
Complete build, light alloy wheels, rim brake, 7-9 speed	~8.5 – 9.5 kg (bare bike)
Complete build, carbon wheels, disc brake, higher-spec groupset	~7.5 – 8.3 kg (bare bike) — but pushes cost toward S\$4,000+, working against your cost target
+ Both child seats fitted	+2.4 – 3.1 kg $\rightarrow$ ~11 – 12.6 kg total

4.3 Tools required (all of Option A's tools, plus)

- Wheel truing stand + spoke key (or outsource wheel-building to a local shop, ~S\$40-60/wheel labour)
- Headset press / headset cup tool (installing headset into a bare frame)
- Crank puller (if not using self-extracting cranks)
- Cable-routing/housing gauge for a frame with unfamiliar internal routing
- Brake bleed kit (if going hydraulic disc)
- Digital caliper (to check clone frame tube diameters before ordering child-seat clamps/adapters)
- Estimated incremental tool cost: S\$150 – 300 on top of Option A's kit

5. Option C — Used Brompton G Line (20") + Titanium/Carbon Modification

The G Line is Brompton's newer (2024-onwards) gravel-oriented model: 20" wheels (vs 16" on classic Brompton), wider gravel tyres, hydraulic disc brakes, and an 8-speed Shimano Alfine internal hub. Stock weight is 13.9-14.8kg — heavier than classic Brompton, but a documented enthusiast build (Ti rear triangle + carbon fork + carbon wheels + exotic groupset) got a G Line down to 8.3kg, proving the mod ceiling is very high if budget allows.

5.1 Bill of materials & cost (SGD equivalent)

Item	Est. cost	Source / notes
Used Brompton G Line, 8-speed, good condition	S\$2,700 – 3,300	New MSRP is ~US\$3,500 (~S\$4,750); a lightly-used example (inspected/tuned) was seen at US\$2,249 (~S\$3,050) against that MSRP — G Line is newer to the used market than classic Brompton so selection is thinner; check Carousell SG and G Line-specific Facebook groups, and UK Gumtree/eBay if travelling
Carbon front fork (e.g. Silverock, disc-brake compatible, 100mm width)	S\$300 – 420	Taobao/AliExpress direct (Silverock official AliExpress store) cheapest; combo fork+rear-triangle sets occasionally discounted to ~S\$220-280 on AliExpress flash sales
Titanium rear triangle (135/130mm width, disc-brake compatible, for 20" wheel)	S\$350 – 480	Taobao/AliExpress direct; G Line-specific width/disc-mount specs make this pricier than classic 16" Brompton Ti rear triangles
Titanium bottom bracket + stem	S\$150 – 220	Taobao/AliExpress
Carbon or Ti seatpost	S\$120 – 180	Taobao/AliExpress
Lighter gravel/slick tyre swap (pair, e.g. Schwalbe-equivalent)	S\$100 – 150	Local shop or Taobao
Titanium bolt kit (full set)	S\$90 – 130	Taobao/AliExpress
Front child seat (Bromptonic/Bobike-style, G Line-compatible)	S\$180 – 260	Taobao direct vs branded EU resellers
Rear seat: DCCH titanium child seat (explicitly listed as A/C/P/T/G-line compatible)	S\$370 – 420	~£220 from Fantastic4Toys (UK); this is the lightest rear-seat option found in this research (~780g total incl. frame) and purpose-fits G Line — worth the premium over a generic rear mount
Consumables (grease, cable, housing, misc.)	S\$50 – 70	Local

Subtotal: roughly S\$4,410 – 5,630 at the high end, but realistically S\$4,100 – 4,700 once you shop Taobao/AliExpress flash sales for the fork+rear-triangle combo set and buy the used G Line at the lower end of its range.

5.2 Weight plan

Configuration	Est. weight
Stock G Line, 8-speed, steel frame/rear frame + aluminium fork	~13.9 – 14.8 kg
After: carbon fork (-0.4kg), Ti rear triangle (-0.5kg), Ti BB (-0.44kg), Ti stem (-0.2kg), carbon seatpost (-0.19kg), lighter tyres (-0.5kg), remove fenders/rack if fitted (-0.85kg each)	~11.5 – 12.5 kg (bare bike, moderate/cost-effective mod level)
Enthusiast extreme build (carbon wheels, SRAM AXS, full carbon cockpit, carbon brakes) — for reference only, not cost-effective	8.3 kg (documented build, cost likely S\$8,000+)
+ Front child seat installed	+1.0 – 1.3 kg
+ DCCH titanium rear child seat installed	+0.78 kg
Both seats fitted, ready to ride (moderate/cost-effective mod)	~13.3 – 14.6 kg total

Note: the G Line starts heavier than classic Brompton, so even after a sensible Ti/carbon mod it will still weigh more than a modded 16" Brompton (Option A) — the trade you're making is weight for bigger wheels, disc brakes, and a more planted ride with kids aboard, not outright lightness.

5.3 Tools required

- Same core kit as Option A (hex/Torx set, torque wrench, pedal spanner, BB tool, tyre levers, cable/housing cutters, Ti-prep anti-seize grease, repair stand)
- Hydraulic disc brake bleed kit + mineral oil or DOT fluid (per G Line's brake spec) if servicing/swapping brakes
- Shimano Alfine hub-specific tools if servicing the internal gear hub (hub spanner, gear indicator alignment)
- Disc rotor truing tool
- Estimated tool cost if starting from scratch: S\$220 – 380 (higher than Option A due to disc brake/hub-specific tools)

6. Front & Rear Child Seat Options (all builds)

Product	Position	Est. price (SGD)	Notes
Genuine-style front seat (Bromptonic-brand, UK-based)	Front, head-tube block mount	~S\$260	Free worldwide shipping, 12-20 days to Asia; for ages ~2-6, up to 125cm
Bobike Mini/One + Milian "Bryan" adapter	Front, handlebar-area mount	~S\$300 – 350 (seat + adapter)	Requires the adapter block; works with most handlebar types
Milian "Pere" rear seat + mount	Rear, seat-tube clamp	~S\$330	Ages ~2.5-5/6; needs a Brompton Pentaclip seat clamp; adds ~3cm protrusion when bike is folded
Fantastic4 (F4) quick-detach child seat	Rear (quick-release, foldable)	~S\$80 (925g)	Cheapest branded option found; shipping origin not disclosed on listing — verify lead time before relying on it for a trip
DCCH titanium child seat (Fantastic4Toys)	Rear, quick-detach	~S\$390 – 420 (780g total)	Lightest rear seat found; explicitly compatible with A/C/P/T/G-line Brompton frames — best pick for Option C, and a premium lightweight pick for Options A/B too if budget allows

Product	Position	Est. price (SGD)	Notes
Taobao-direct equivalents (search "小布儿童座椅" / "折叠车儿童座椅")	Front or rear	~S\$60 – 150 each	Same factories supply several Western-branded resellers at 2-3x markup; biggest single cost-saving lever in this whole plan — worth sourcing both seats this way

7. Where to Buy Cheapest (incl. China) & Shipping-to-Hotel Logistics

7.1 Recommended sourcing by item

- Base used Brompton (classic or G Line): Carousell (Singapore) first — zero shipping/customs friction, easy to inspect hinge wear/disc rotors in person. Only consider UK/Japan used listings (Gumtree, Preloved, Mercari, Yahoo! Auctions Japan) if you're already travelling there and can fold it into checked luggage — shipping a full bicycle internationally usually costs more than the savings. G Line's used market is thinner than classic Brompton's, so widen your search to include G Line owner groups and eBay UK/US.
- Titanium/carbon frame parts (fork, rear triangle, full frame kits): Taobao direct is consistently the cheapest — often half to a third of the price of the same part relisted on eBay/AliExpress by a Western reseller (e.g. EasyTi-branded parts on eBay run noticeably higher than the same factory's Taobao listings; Silverock's own AliExpress store is a good middle ground for G Line-specific disc-brake-width parts). You will need a China forwarding/agent service (e.g. Superbuy, CSSBuy, Wegobuy, Sifubuy) since Taobao does not ship internationally on its own.
- Pre-built titanium clone complete bikes (Option B): tibicycles.com is Singapore-based (TaiSeng MRT) — convenient, no import hassle, but 2-4x the cost of self-sourcing parts. Use this only if you want a turnkey Option B bike and are willing to pay the assembly/convenience premium.
- Child seats: Taobao-direct is again cheapest for generic seats; the DCCH titanium seat (Option C's rear-seat pick) is currently only found via Fantastic4Toys (UK) — worth checking whether the same factory also lists it on Taobao/AliExpress before buying the UK-branded version at full price.
- Tools: generic hex/torque wrench sets are cheapest on Taobao/AliExpress; disc-brake bleed kits and Alfine hub tools (needed for Option C) are also cheaper sourced from Taobao than from Shimano-branded Western retailers.

7.2 Shipping to an overseas hotel while travelling

- Use a single China forwarding agent to consolidate multiple Taobao purchases (frame, fork, rear triangle, both seats, bolt kit) into one parcel — cuts per-shipment freight fees significantly versus shipping each item separately.
- Confirm with the hotel in advance (email/WhatsApp) that they accept package holding for an in-house guest, and address the parcel exactly as: your full name, "Guest arriving [date], reservation #[xxxx]", hotel name and address, hotel front-desk phone number.
- Build in a buffer: have the agent ship so the parcel arrives 3-5 days before your check-in, and get tracking with signature-on-delivery where possible so it isn't sitting unclaimed at the front desk too long.
- For bulky items (full frame kit, wheels), check the hotel's package size policy — some business hotels cap parcel size/weight they'll hold for guests.
- Carry the assembled/folded bike home in checked luggage or as an oversized item rather than shipping the finished bike — shipping a fully built bike internationally is expensive and risks transit damage; shipping loose parts and assembling at home (or a local shop) is both cheaper and safer.
- As requested, this plan excludes Singapore GST import cost and tourist-scheme tax rebates from all figures above — factor Singapore's prevailing import GST rate separately when the parcels actually land, since it will add to the landed cost.

8. Key Risks & Caveats

- IP/trademark: Brompton's original folding-hinge patents have expired, which is why clone frames (from Hexati, EasyTi, Pacific/Pikes-style factories, etc.) are widely available, but the Brompton brand/logo itself is still trademarked — buy "Brompton-compatible"/"trifold" clone frames, not counterfeit-branded ones.

- Clone frame tolerances vary by factory batch — order a digital caliper reading of head-tube and seat-tube diameters from the seller before committing to a child-seat mount purchase, since seat/adaptor fit is the single biggest risk in Option B.
- Titanium-to-steel/alloy interfaces need anti-seize/Ti-prep grease to avoid galling; skipping this is the most common DIY mistake reported in the forums researched.
- A modified used Brompton or G Line (Options A/C) retains most of its resale value since it's still a genuine frame; a clone-frame Option B build has little to no resale market — treat it as a build-to-keep, not a build-to-flip.
- Inspect any used Brompton's/G Line's frame hinge and main hinge clamp for play/cracking before buying — this is the highest-wear part on high-mileage units and the costliest to replace. For G Line specifically, also check disc rotor wear and Alfine hub oil-service history.
- G Line-specific Ti/carbon aftermarket parts are a newer, smaller niche than classic Brompton parts — expect fewer sellers, longer lead times, and less price competition than for 16" Brompton Ti parts; buy from established Taobao/AliExpress stores with good review history rather than the cheapest unknown listing.

9. Suggested Next Steps

- 1) Decide between Option A (cheapest, lightest-per-dollar, best for smooth city riding) and Option C (costlier, more planted/confidence-inspiring for carrying two kids on rougher surfaces) before shopping — this determines which used bike you hunt for.
- 2) Start watching Carousell for a well-maintained 3-speed steel Brompton (Option A, budget S\$650-950) or a used G Line 8-speed (Option C, budget S\$2,700-3,300); widen to UK/overseas listings only if you're travelling there soon.
- 3) In parallel, identify a China forwarding agent and get comfortable with the Taobao/AliExpress purchase flow before your next work trip.
- 4) Order the fork + rear triangle + bolt kit + both child seats as one consolidated shipment, timed to land at your destination hotel 3-5 days before a planned trip, or simply to your home/office if no trip is imminent.
- 5) Assemble your chosen build first and live with it for a season to validate both child seats fit your kids and your routine before considering a second bike.
- 6) If you still want a second, ultralight bike afterward, revisit Option B as a dedicated solo/commuter build once you've got hands-on experience from Option A or C.